Nguyen Duong Hieu

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Professional Summary

A dedicated Data Science student with a strong research interest in Causal Inference and its intersection with Machine Learning, particularly in causal representation learning, out-of-distribution (OOD) generalization, and trustworthy AI.

Research interests: Causal inference and causal machine learning; causal representation learning; out-of-distribution generalization; trustworthy and probabilistic machine learning.

Education

National Economics University, BS in Data Science in Economics and Business — Sept 2023 – Present

- GPA: 3.79/4.0
- Relevant coursework: Machine Learning, Probability & Statistics, Optimization, Linear Algebra, Data Structures & Algorithms

Tran Phu High School for the Gifted, Natural Science 1 — 2020 – 2023

Experience

AI Intern, Fintech & Payment, VinSmart Future (Vingroup) — July 2026 – Present

- Developed an uplift-modeling proof of concept for targeted fintech marketing by estimating Conditional Average Treatment Effects (CATE) rather than conventional response probabilities.
- Built an experimentation pipeline on the Hillstrom and Criteo benchmarks, covering preprocessing, treatment-effect estimation, policy evaluation, and cost-effectiveness analysis under budget constraints.
- Designed a simulated A/B testing framework to evaluate whether learned targeting policies produce incremental outcomes.

Student Researcher, DAAI Lab – National Economics University — Jan 2026 – Present

- Developed a multimodal workflow combining OCR and PhoWhisper ASR with Gemini for structured entity extraction from noisy visual and acoustic inputs.
- Built hybrid retrieval with Sentence-BERT, BM25, and Reciprocal Rank Fusion over a 50,000-SKU enterprise catalog.
- Achieved F1-scores of 0.91 on visual inputs and 0.85 on acoustic inputs, with Recall@5 of 0.93 under noisy conditions.

Research Assistant, "Does Network Position Cause Scholarly Success?" — Spring 2026

With Prof. Mike Nguyen (University of Southern California) and Prof. Minh Nguyen (Florida Atlantic University)

- Curated and structured large-scale academic collaboration networks to support causal analysis of how network position affects scholarly productivity.

Student Researcher, DATAOPT Lab – National Economics University — July 2024 – Jan 2026

- Implemented mathematical and convex optimization algorithms, reproduced experiments from the literature, and evaluated their behavior in machine-learning settings.

Teaching Assistant, Python for Data Science & Power BI – National Economics University — Aug 2025 – Dec 2025

- Prepared practical exercises, explained Python and data-analysis concepts, and provided structured feedback on student submissions.

Publication

Promoting circular consumption behavior: Factors influencing Vietnamese Generation Z's intention to extend the lifespan of personal electronic devices — 2026

Le Thu Thao, ***Nguyen Duong Hieu****, Pham Phuong Trinh, Nguyen Minh Tung*

22nd International Conference on Socio-economic and Environmental Issues in Development (ICSEED 2026)

- Applied PLS-SEM to 733 respondents using an integrated Theory of Planned Behavior and Push-Pull-Mooring framework.

Projects

Hanoi Temperature Forecasting System [GitHub] — Late 2025

- Built a five-day forecasting pipeline from more than ten years of weather data, engineering 90+ lag, STL decomposition, and FFT features.
- Trained and tuned a multi-output CatBoost model ($R^2 \approx 0.84$), then evaluated predictions with SHAP explanations

and PSI/KS-test drift monitoring.

DentFlow AI – AI-Assisted Dental Booking & Clinic Flow Platform [GitHub] [Live Demo] June – July 2026

- Built guarded AI-assisted intake and adaptive booking with structured outputs, deterministic fallbacks, and conflict-safe slot allocation.
- Implemented deterministic clinic-flow optimization with room compatibility scoring, service queues, route sequencing, realtime updates, and role-based access.

Fresh vs. Spoiled Fruit Classification: Leakage-Aware CNN Evaluation [GitHub] Fall 2026

- Audited a 16,000-image dataset, removing 1,273 duplicates and building a perceptual-hash grouped split to prevent the near-duplicate leakage that would affect 6.5% of a random test split.
- Compared a baseline CNN, a multi-task residual CNN, and fine-tuned MobileNetV2 (99.73% joint accuracy on 2,203 test images), using augmentation and fine-tuning-depth ablations plus Grad-CAM, which revealed background reliance in the smaller models.

Technical Skills

- **Research methods:** Causal inference, CATE/uplift modeling, DAGs, experimental design, probabilistic ML, optimization, statistical analysis
- **Programming and data:** Python, SQL, TypeScript, Pandas, NumPy, Scikit-learn
- **Machine learning:** TensorFlow/Keras, CatBoost, Optuna, SHAP, Sentence-BERT, BM25, LLMs, NLP, computer vision, OCR/ASR pipelines
- **Engineering tools:** Git, Docker, GitHub Actions, Next.js, React, Supabase, Streamlit, Power BI, LaTeX
- **Domain knowledge:** Microeconomics and macroeconomics fundamentals

Selected Achievements

- **2026** Third Prize, University-Level Scientific Research Competition, National Economics University
- **2025** Top 10, Hanoi Mathematical Modeling Competition
- **2024** Academic Encouragement Scholarship (Excellent), National Economics University
- **2024** Outstanding Youth Union Member Award, National Economics University
- **2023** Consolation Prize, Hai Phong City Mathematics Competition
- **2022** IELTS 7.0